

### Prob 1 - Ramp input

$$r(t) = \begin{cases} t, & t \leq 1 \\ 1, & t > 1 \end{cases}$$

$$u(t-1) = \begin{cases} 0, & t < 1 \\ 1, & t > 1 \end{cases}$$

a)  $r(t)$  can also be written as

$$r(t) = t(1 - u(t-1)) + u(t-1) = t - t u(t-1) + u(t-1)$$

b)  $R(s) = \mathcal{L}(t) - \mathcal{L}(t \cdot u(t-1)) + \mathcal{L}(u(t-1))$

$$= \frac{1}{s^2} - e^{-s} \mathcal{L}(t+1) + e^{-s} \mathcal{L}(1)$$

$$= \frac{1}{s^2} - \frac{e^{-s}}{s^2} - \frac{e^{-s}}{s} + \frac{e^{-s}}{s}$$

$$= \underline{\underline{\frac{1}{s^2} (1 - e^{-s})}}$$

c)  $y'' + y = r(t)$

$$s^2 F(s) + F(s) = \frac{1}{s^2} (1 - e^{-s})$$

$$F(s)(s^2 + 1) = \frac{1}{s^2} (1 - e^{-s})$$

$$F(s) = \frac{1}{(s^2 + 1)} \left( \frac{1}{s^2} - \frac{e^{-s}}{s^2} \right) = \frac{1}{(s^2 + 1)s^2} - \frac{e^{-s}}{(s^2 + 1)s^2}$$

$$F(s) = \mathcal{L}(\sin(t) * t) - \mathcal{L}(u(t-1)(\sin(t) * t))$$

$$\sin(t) * t = \int_0^t \sin(y)(t-y) dy = t - \sin(t)$$

$$f(t) = t - \sin(t) - u(t-1)(t-1 - \sin(t-1))$$

### Prob 2 - Dirac input

a)  $y'' + y = \delta(t)$

no in. cond.:  $s^2 F(s) - \cancel{s F(0)} - \cancel{F'(0)} + F(s) = e^{-s \cdot 0} = 1$

$$F(s) = \frac{1}{(s^2 + 1)}$$

$$f(t) = \sin(t)$$

with in. cond.:  $s^2 F(s) - \overset{=0}{\cancel{s F(0)}} - \overset{=1}{\cancel{F'(0)}} + F(s) = 0$

$$F(s) = \frac{1}{(s^2 + 1)}$$

$$f(t) = \sin(t)$$

$$b) \quad y''' + y = 0 \quad , \quad y(0) = y'(0) = 0 \quad , \quad y''(0) = 1$$

$$s^3 F(s) - \overset{=0}{s^2} F(0) - \overset{=0}{s} F'(0) - \overset{=1}{F''(0)} + F(s) = 0$$

$$F(s) (s^3 + 1) = 1$$

$$F(s) = \frac{1}{(s^3 + 1)} = \frac{1}{(s+1)(s^2 - s + 1)} = \frac{A}{(s+1)} + \frac{Bs}{(s^2 - s + 1)}$$